Scramble

AI-Powered Civilian Evacuation Intelligence

For Smart Cities

SMART CITIES TRACK • AI-POWERED SOLUTIONS

13M

People in the city

54

Years since last drill

0

Dynamic evacuation systems

May 2025

Operation — gap confirmed

THE PROBLEM

What happens when the sirens stop?

"On Jan 23rd, 2026, sirens rang across Texas. 230 million people heard them. None knew what to do next."

ROUTING GAP

Everyone told to go to nearest shelter → gridlock in 8 mins → shelter overflows in 20 mins

LAST-MILE GAP

Alerts in English on apps needing internet — invisible to 60% of Texas population

COORDINATION GAP

NDRF, BBMP, Police, Hospitals — all operating in silos, no shared picture

USA has invested ₹48,000 Cr in Smart City infrastructure — but has NO dynamic civilian evacuation system

HOW IS Scramble DIFFERENT?

	EXISTING SYSTEMS	SCRAMBLE
Instruction type	✗ Broadcast to everyone	✓ **Personalised per zone + route + ETA**
Hazard data	✗ Static maps, last year's data	✓ **Live Sentinel-1 SAR satellite feed**
Scale handling	✗ Same route → instant gridlock	✓ **AI flow-balances across city capacity**
Shelter awareness	✗ None	✓ **Real-time occupancy, auto-reroutes**
Languages	✗ English only	✓ **Kannada / Hindi / Tamil / Telugu / EN**
Offline access	✗ Needs internet	✓ **SMS fallback — works on 2G/no signal**
Informal settlements	✗ Invisible	✓ **Community-mapped, fully included**
AI hardware	✗ N/A	✓ **AMD ROCm — open, fast, India-deployable**

UNIQUE SELLING PROPOSITION

**Evacuation is a Flow Optimization Problem.
Nobody has solved it. Until now.**

AI COORDINATES, NOT JUST NOTIFIES

Routes 230 M people optimally across shelters in real-time. Like Google Maps — for evacuation.

SATELLITE-DRIVEN LIVE HAZARD MAP

Sentinel-1 SAR detects floods through clouds. Blocked roads auto-removed from routes instantly.

OFFLINE-FIRST INDIA DESIGN

SMS fallback. 2G-compatible. Kannada/Hindi/English alerts. Built for any infrastructure.

FEATURES

8 capabilities across 4 layers

Live Flood Detection

Sentinel-1 SAR satellite auto-maps hazard zones

Dynamic Routing Engine

AI assigns zone→shelter→route balanced across city

Personalized Alerts

Exact instructions per citizen: route, ETA, what to bring

5-Language LLM Alerts

Kannada, Hindi, Tamil Telugu, English via Llama 3

SMS Offline Fallback

Works without internet on 2G — India-first design

Community Mapping

Informal settlements mapped by local volunteers

Responder Dashboard

Live movement, shelter load bottleneck alerts for NDRF

Family Check-In

Mark safe, track family status during evacuation

PROCESS FLOW

From disaster detection to citizen safety

1

Satellite Detects Hazard

Sentinel-1 SAR
Google Earth Engine

2

Hazard Map Updated

Flooded roads
auto-blocked

3

AI Routing Engine Runs

Zone → shelter
assignments optimised

4

Citizens Get Alert

App + SMS
5 languages

5

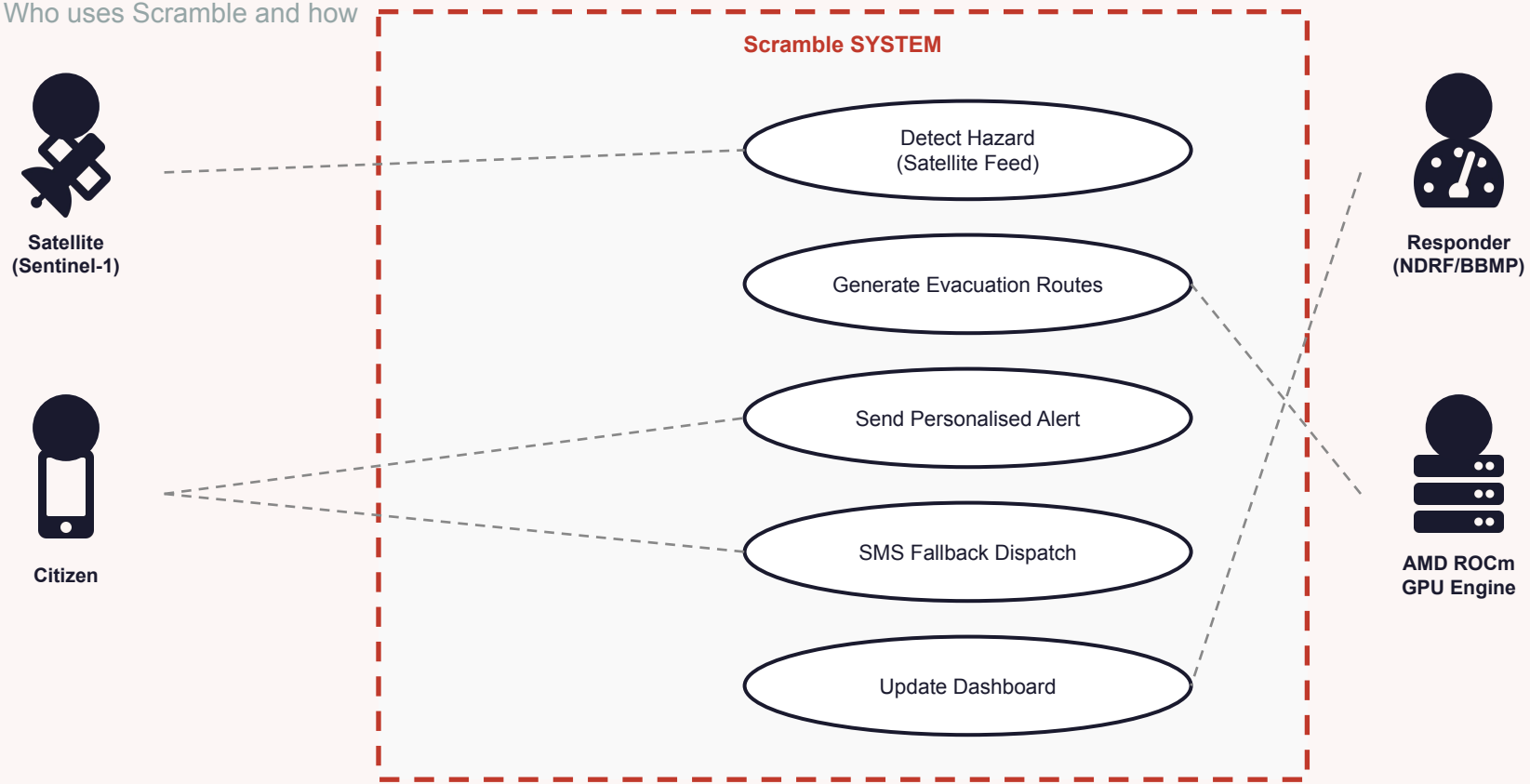
Responders Dashboard

Live map
resource coordination

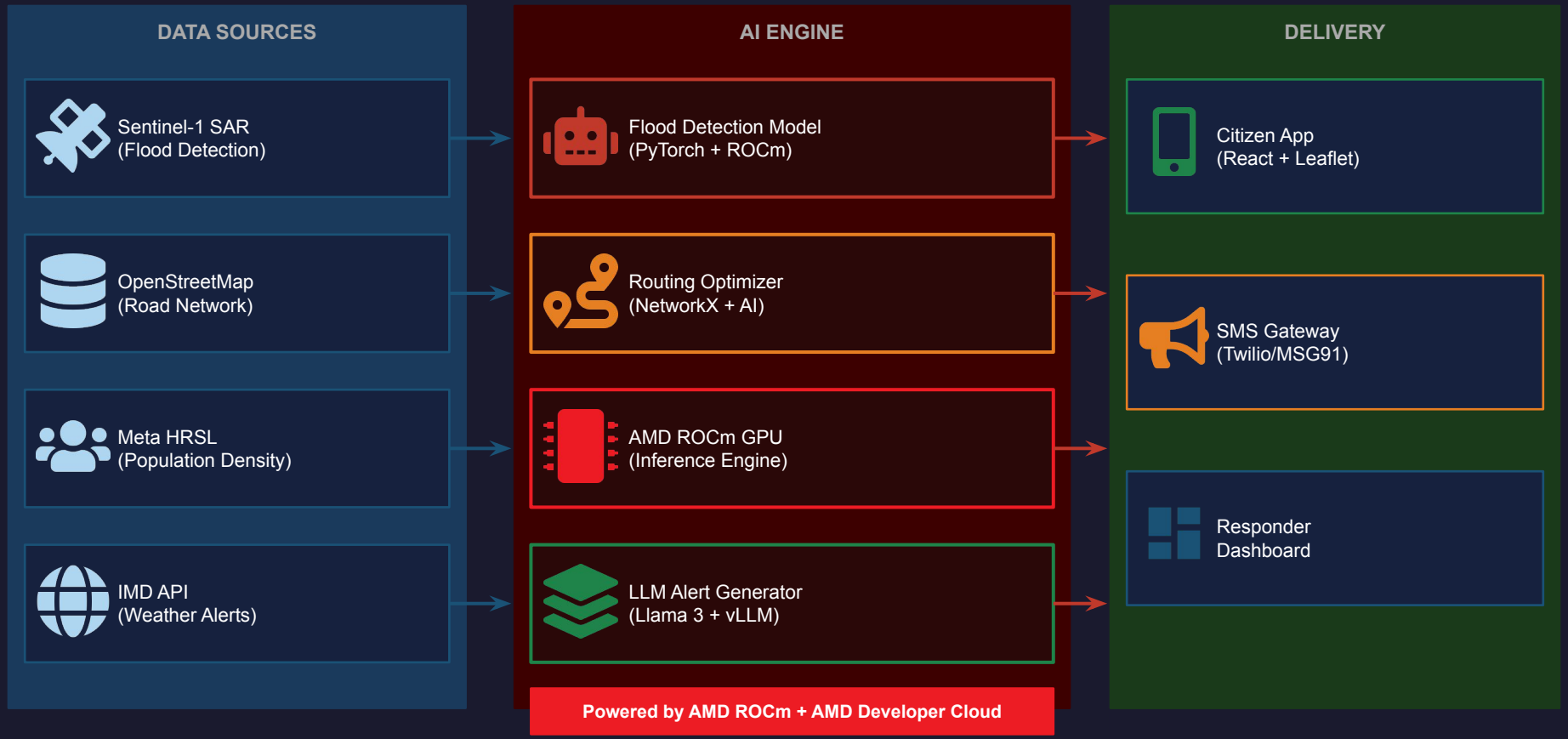
⚡ Total time from satellite detection to citizen SMS: under 3 minutes | Runs on AMD ROCm GPU infrastructure

USE-CASE DIAGRAM

Who uses Scramble and how

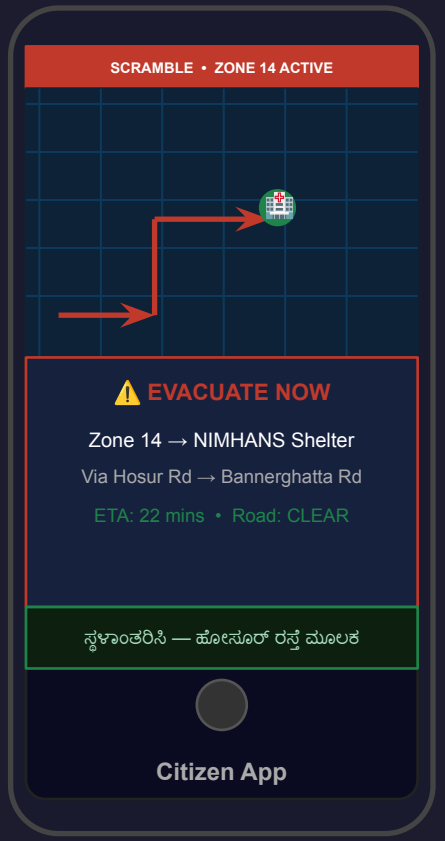


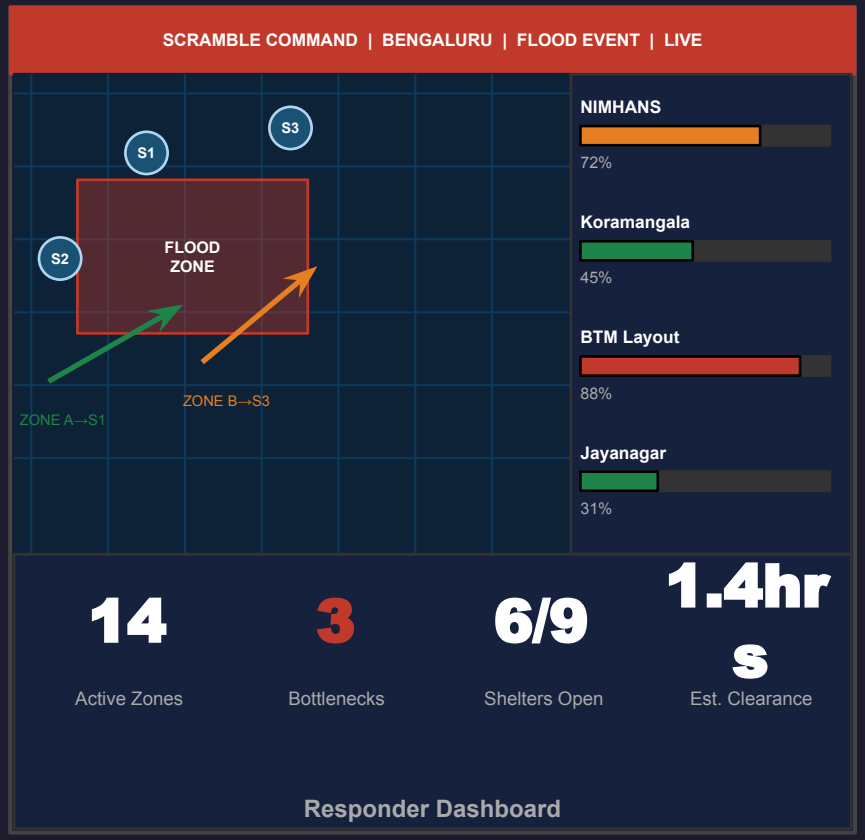
ARCHITECTURE DIAGRAM

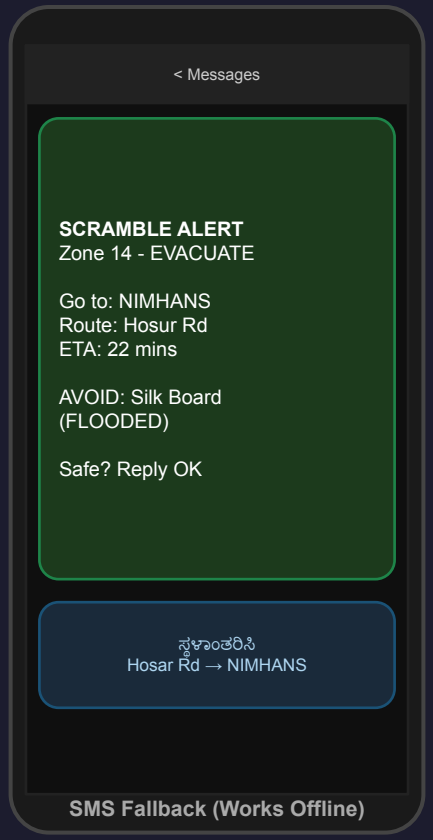


WIREFRAMES

Citizen App • Responder Dashboard • Alert SMS







AMD PRODUCTS

AMD

AMD ROCm 7

Open GPU Computing Platform

- Train Sentinel-1 flood detection model (PyTorch)
- Run inference — flooded zone detection in <1s
- 3.5× faster than previous ROCm versions

AMD

Llama 3 + vLLM on ROCm

Open LLM via AMD GPU Stack

- Generate multilingual alerts — Kannada, Hindi, Tamil, Telugu, English
- Alert generated in under 2 seconds per zone
- Fully open-source — no API cost

AMD

AMD Developer Cloud

Free GPU Credits for Students

- Full scramble AI pipeline hosted free for prototype & demo
- ROCm-native environment — one-click setup
- Access at amd.com/developer

AMD ROCm is fully open-source • No NVIDIA/CUDA lock-in • India-deployable • Free student access via AMD Developer Cloud

TECHNOLOGIES

Full open-source stack — learnable in 2 months

SATELLITE	**Google Earth Engine** Sentinel-1 SAR flood detection	**Meta HRSL** Population density grid	**OpenStreetMap** Road network graph
AI / ML	**PyTorch + ROCm** Flood detection model (AMD GPU)	**NetworkX + A*** Graph routing optimizer	**Llama 3 + vLLM** Multilingual LLM (AMD GPU)
BACKEND	**Python + FastAPI** Core API + routing engine	**PostgreSQL + PostGIS** Geospatial database	**Twilio/MSG91** SMS offline dispatch
FRONTEND	**React + Leaflet.js** Citizen app + map UI	**Tailwind CSS** Responsive UI	**WebSockets** Real-time dashboard

✓ **All data sources free • All AI frameworks open-source • AMD Developer Cloud: free GPU credits for students**

ESTIMATED IMPLEMENTATION COST

PROTOTYPE

Now → April 27

₹0

Zero cost

• AMD Developer Cloud	Free (student credits)
• Google Earth Engine	Free (academic)
• OpenStreetMap / Meta HRSL	Free
• Twilio SMS (dev tier)	Free up to 100 SMS
• All frameworks	Open-source

PILOT

1 City, 3 Months

₹8–12L

Per city pilot

• AMD Cloud GPU (24/7)	~₹3L/month
• SMS gateway (100K msgs)	~₹1.5L/month
• Dev team (3 engineers)	~₹2L/month
• Government API access	₹0–1L
• Server + DevOps	~₹1L/month

CITY SCALE

Full Bangalore Deploy

₹2–4Cr

One-time + ₹50L/yr

• AMD Instinct GPU cluster	₹1–2Cr one-time
• Full engineering team	₹80L/yr
• City integration + APIs	₹50L one-time
• Community mapping prog.	₹30L/yr
• BBMP license revenue	Covers running costs

"The sirens rang.
230 million people heard them.
None knew what to do next."

That is not an alert problem.

That is an evacuation intelligence problem.

That is what scramble solves.